

Rahul Rajan

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SUMMARY

Computer Science and Engineering undergraduate with experience in software development, machine learning, reinforcement learning, and algorithmic problem solving. Experienced in full-stack and mobile development, with research work in WBANs, IoMT, and Q-learning-based adaptive routing. Skilled in Python, MATLAB, web technologies, and network simulation, with interests in AI/ML, Reinforcement Learning, intelligent systems, and distributed networks.

EDUCATION

B.Tech - Computer Science and Engineering, Amrita Vishwa Vidyapeetham CGPA - 7.05	2023 – present
International Indian School, Dammam, Saudi Arabia Class 10 - 81.4% Class 12 - 82.6%	2020 – 2022

PROJECTS

Q-Learning Based Adaptive Routing and Priority Scheduling for WBANs, <i>MATLAB, Q-Learning, Reinforcement Learning, Wireless Body Area Networks</i> - Developed a Q-Learning-based adaptive routing framework for WBANs, dynamically selecting energy-efficient and reliable next-hop nodes based on network state, connectivity, and transmission conditions - Integrated priority-aware scheduling to differentiate critical medical data from routine sensor data, prioritizing time-sensitive information through low-delay and reliable routing decisions. - Evaluated single-hop, two-hop, and hybrid routing strategies using packet delivery ratio, energy consumption, end-to-end delay, throughput, and network lifetime.	2026
Stride Wars - Location Based Fitness Gamification Platform, <i>React.js, Node.js, Express.js, MongoDB, REST APIs, JWT, GPS/Map APIs</i> - Developed a location-based fitness platform that gamifies running by converting GPS-tracked routes into interactive territories for users to capture, conquer, and defend. - Implemented secure user authentication and authorization using JWT and bcrypt, with regional/global leaderboards, achievements, and competitive gameplay features. - Designed a three-tier MERN architecture with React.js, Node.js/Express.js, and MongoDB, integrating REST APIs and GPS/Map APIs for real-time route tracking and territory visualization.	2025
GEOSAFE - Location Based Community Safety App, <i>Kotlin, Android, OpenStreetMap, OSMDroid/MapLibre, SQLite/Room, MVVM</i> - Developed a location-based Android application enabling users to report nearby crime incidents and visualise reported activity through an interactive OpenStreetMap-based interface. - Implemented real-time location tracking, risk-zone detection, and in-app alerts to notify users when entering high-risk areas, along with density-based crime heatmaps. - Designed an MVVM-based architecture with local SQLite/Room storage for incident management, supporting user reporting, map visualization, and location-aware safety features.	2025

TECHNICAL SKILLS

Languages, <i>Python, C, Java, JavaScript, Kotlin, SQL</i>
Frameworks & Tools, <i>PyTorch, TensorFlow, React.js, Node.js, Express.js, Laravel, MATLAB, NS-3, MongoDB, MySQL, PostgreSQL, Git, GitHub, REST APIs, Tailwind CSS</i>
Concepts, <i>Reinforcement Learning, Q-Learning, Machine Learning, Deep Learning, Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Network Simulation, Cloud Computing, Web Development</i>

RESEARCH INTEREST

Reinforcement Learning for Intelligent Medical IoT (IoMT) Exploring reinforcement learning-based routing, resource optimization, and priority-aware communication in Wireless Body Area Networks (WBANs), with a focus on improving energy efficiency, reliability, latency, and Quality of Service (QoS) for critical medical data.
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SOCIAL OUTREACH

Vector-Borne Disease Awareness Program — St. John's HSS, Mattom, Mavelikara - Led a team initiative educating 50+ students on vector-borne disease prevention, identifying monsoon-related health risks as a potential barrier to academic progress. - Organised and delivered an interactive awareness session, coordinating with the team and encouraging students to take a preventive-health pledge and spread awareness, demonstrating leadership, communication, and social responsibility.
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